

do not actually compete—the purpose of a quantum computer is not simply to solve classical computing problems faster than a supercomputer. It is fundamentally different, and ideally suited to crunch complex problems that require calculating a large number of possible combinations.

The problems best left to quantum computers are known as bounded-error quantum polynomials (BQP), that is, problems that can be solved by a quantum computer in polynomial time, such as factoring, searching an unordered database, and solving systems of linear equations. Problems like this abound in fields like optimisation, materials science, cryptography, artificial intelligence, machine learning, sensing, telecommunications, and finance. Many processes in nature also involve many-body quantum interactions. This makes quantum computing an essential tech to simulate natural processes like protein folding, molecular formations, photosynthesis, and superconductivity, and use this understanding to develop better nature-inspired technologies.

“Today I cannot see that there will be something like a quantum PC,” quips Professor Rainer Dumke, Principal Investigator at the Centre for Quantum Technologies, Professor at Nanyang Technological University, Singapore, and Lead for hardware at Singapore’s National Quantum Computing Hub (NQCH). “We will witness in the future that more and more data centres will have a quantum computer on the premises. To give a meaningful timeframe for this is very challenging, especially at this early stage. However, we see already that supercomputers and data centres around the globe are adopting quantum computers of the current generation.”

At the NQCH, they are working to establish a Singapore-built quantum computer, which can be accessed by users remotely. They also hope to develop an advanced quantum processor test platform, along with various components like chips, fast

Horizons of quantum growth

Horizon 1, also called now or near term, includes transactional use cases, such as credit scoring, vehicle routing, chemical design, chemistry, and drug/protein structure prediction.

Horizon 2, also called near term, follows with oil processing and shipping, refining processes, drilling, livestock, disruption management and supply chain issues, investment risk analysis, clinical trial acceleration, and optimisation of manufacturing and fabrication.

Horizon 3, also called future looking, consists of seismic imaging, consumer recommendation with financial analysis, disease risk prediction, and structural design for buildings.

(Credits: Paul Nashawaty, Senior Analyst at Enterprise Strategy Group, a division of TechTarget)

electronics, and cryoelectronics for the next generation of superconducting quantum processors.

Overcoming the hardware challenges

In an actual quantum computer, physical systems, such as the spin of an electron or the orientation of a photon, are used to store qubits. These, as you can imagine, are pretty fragile systems, and are easily affected by various factors, such as disturbances in the earth’s magnetic field, radiation from devices in the vicinity, heat, neighbouring qubits, and so on. These factors, which affect the information in a qubit, are known as ‘noise,’ and since today’s quantum computers are prone to their effects, they are known as noisy intermediate-scale quantum (NISQ) devices.

Unfortunately, even reading or accessing a qubit amounts to noise! Noise can cause the information in the qubits to change, or even fade away! This is known as decoherence—that is, the information in the qubit is no longer coherent. Noise and resulting decoherence are two of the biggest hurdles in the way of quantum computers.

Quantum computer makers try to protect the qubits from noise by using cooled silos, controlled pulses of energy, etc. So, if you look at a quantum processor, such as the one from IBM, the wafer is just about the size of the one in your laptop, but it is housed in

a hardware system the size of a car, made up of cooling systems to keep the superconducting processor at its ultra-cold operational temperature.

The task of mitigating the effect of noise is known as quantum error correction. Mostly, a lot of spare physical qubits are used to store every logical qubit, so that in case one loses its coherence you can verify with another. Researchers also use noise simulations to try and understand the nature and effect of noise, and develop noise-tolerant algorithms, which will give fairly good results despite the noise. Finnish company Algorithmiq is working to develop noise-resilient quantum algorithms for drug discovery. Their strategy involves using standard computers to ‘un-noise’ quantum computers!

“Techniques for error mitigation and error correction are likely to be the way forward on NISQ computers. There is currently a trade-off between the number of qubits we can use and the depth of the quantum circuit. The product of the two is referred to as the quantum volume. Researchers are finding ways to increase the quantum volume, and demonstrate the advantage of quantum algorithms. One such approach is hybrid quantum computing, where the computational task is shared between classical and quantum processors,” says Prof. Prabhakar.

“The research on quantum fault tolerance is ongoing, and the initial